Nicola Calzone

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Job Experience

AlfaGroup, *Machine Learning Engineer, Project Contract* Sep 2026 – Present
- Developing ML models for the low-latency clustering and detection of fraudulent or suspicious transactions.
- Pre-training generative models on the company's GPU infrastructure to generate millions of anonymized, realistic synthetic transactions from real transaction data.

LAIFE Reply, *Full Stack AI Engineer, Project Contract* Dec 2025 – Sep 2026
- Full-stack architect (Frontend: React.js / Backend: Java and SpringBoot for microservices) for 3-person squad (PM + Mobile dev + me) on a Healthcare application; I own the roadmap and end-to-end delivery of hybrid-search RAG pipeline: indexed 20GB of documentation into LanceDB with full-text + dense embeddings, exposing semantic, corpus and summarization tools with MCP Server. Chatbot for querying DBs and generating graphs.

Charles University, *Student Researcher Assistant, Part-Time* Sep 2024 – Jul 2025
- Co-authored research on Transformers pre-trained on formal grammars (Prof. Bojar, Prof. Satta); under review
- Owned ML experiments end-to-end, from design through iteration: built an encoder-only (BERT-like) acceptor model in PyTorch; trained, evaluated, and scored a seq2seq Transformer translator with Fairseq (96.7% and 74%)

Unicredit, *Back-end Software Developer, Full-Time (during studies)* Feb 2024 – Sep 2024
- Resolved JVM crash under load, resulting in 10% increase peak-hour traffic and full system stability.
- Improved security: eliminated path traversal, XML and HTML injection attacks; raised test coverage to 100%, driving 30% higher bug detection rate. Used Spring Boot, JPA, Jenkins, Bitbucket, Dynatrace, Splunk.

Projects

(1) **BookShare** - I developed a webapp and mobile application initially designed for managing book collections among friends and family, now evolving into a location-based social network for local book exchanges.
(2) **LocalTable** - working on a mobile application connecting local producers of fruits, vegetables, and other natural products directly to nearby businesses (restaurants, bars, cafés) and customers.
(3) Robot navigation, object recognition and pick-and-place through ROS, Gazebo [GitHub],
(4) Open-Source contribution to Natural Language Toolkit to generate and parse SCFGs [link GitHub].

Competitions: Reply AI Challenge 2026 on smart fraud detection; Kaggle Agents Competition: Smart Booking Multi-Agent for restaurants, saloons, etc. [link GitHub]; Deep Learning course competitions [link GitHub]

Education

M.Eng. in Computer Engineering, *Universita degli Studi di Padova, Italy* 2022-25 GPA: 3.88/4.0
M.Eng. Research Thesis, Erasmus+, *Univerzita Karlova, Prague, Czechia* 2024-25 GPA: 4.0/4.0
- Thesis: "Can Transformers Learn SCFGs? An Analysis of Pretraining on Automatically Generated Datasets"
- (1) Acceptor experiment using an Encoder-Only Transformer (BERT-like) with torch to build the model;
- (2) Translator experiment using seq-2-seq Transformer with Fairseq for training, inference and scoring.

High School Diploma, *Liceo Classico P. Giannone, Italy*, GPA: 3.7/4.0
- Honorable mention in the Greek Translation Olympiad and participated twice in the Latin Translation Olympiad

Skills

Languages	Italian (Native), English (C1), Spanish (C1), Russian (B2)
Programming	Python (torch, HuggingFace, pandas, FairSeq), Java (Spring Boot, Maven), C, C++ (ROS, OpenCV, Make), SQL, Assembly (MIPS, x86), Matlab, Docker, PostgreSQL
Deployment	CI/CD (Git, GitHub Actions, Jenkins), Google Cloud (Vertex AI), AWS (Bedrock, EC2, S3)
AI & Data	LLMs, Agentic AI (LangChain, MCP, Google ADK), Vector DBs (LanceDB, FAISS), Cloud (GCP, AWS)
Soft Skills	Problem-solving, mentorship and leadership (built and ran a tutoring practice)

Certificates

NVIDIA: Fundamentals of Deep Learning for Computer Vision , Fundamentals of Deep Learning for Multiple Data Types , Fundamentals of Deep Learning for Natural Language Processing; **Kaggle/Google:** AI agents course; **GitHub:** GitHub Foundations; **Anthropic:** Introduction to Model Context Protocol; MCP: Advanced Topics; Introduction to agent skills; Introduction to subagents; Claude Code 101;

Additional Experience

Erasmus Student Network, *Volunteer* Feb 2019–Jan 2020
- Led groups of 100+ international students, organized tours, finding housing.

Freelance Private Teacher, *Tutoring for Calculus, Physics, Programming* Mar 2022–Present
- Built and ran a solo tutoring practice for calculus, Python programming and Italian language: acquired and retained 30+ clients over 3+ years; 90% success rate in university exams and language certifications.